

Kavya D Bandaranayaka

Edmonton, AB | (780) 233 2625 | kavyadbandaranayaka@gmail.com | [Portfolio](#) | [Github](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Software Developer with 3 years of experience designing and developing enterprise web applications, AI-enabled solutions, REST APIs, and operational systems using C#, ASP.NET Core, SQL Server, React/TypeScript, and Azure. Experienced in building and integrating AI-driven features including RAG pipelines, LLM API integration, prompt engineering, and responsible AI output validation into full-stack applications. Strong background in object-oriented design, unit testing, CI/CD pipelines, containerized deployments, and Tier 2 production support for business-critical applications. Proven ability to translate evolving business requirements into scalable, user-centered digital solutions within Agile/Scrum environments.

TECHNICAL SKILLS

Languages: C#, ASP.NET Core, .NET Framework, JavaScript, TypeScript, SQL, Python

Frontend: React, Angular, HTML5, CSS3, SCSS, Bootstrap, SPA Architecture

Backend: ASP.NET Core Web API, RESTful APIs, Microservices Architecture, Node.js, Express.js, SignalR, API Integration

Databases: SQL Server, PostgreSQL, MongoDB

Cloud & DevOps: Microsoft Azure, Azure DevOps, Azure DevOps Pipelines, CI/CD Pipelines, Docker Containers, GitHub Actions, GitHub Copilot, Azure Functions, Azure Data Factory (Foundational Knowledge)

AI & Data: OpenAI API, pgvector, RAG / LLM Integration

Testing & Core Concepts: Unit Testing, Automated Testing, Object-Oriented Programming (OOP), SOLID Principles, SDLC, Agile/Scrum, Debugging, Code Reviews, Troubleshooting

Enterprise & Operational Systems: Enterprise Application Development, Production Support, System Documentation, Operational Support, System Integrations

Tools: Git, GitHub, GitLab, Visual Studio, Postman, Jira, ServiceNow, SAP

GIS & Analytics: Leaflet, GeoJSON, Python ETL Pipelines, Data Validation, Reporting Workflows

PROJECTS

[AI Document O&A Assistant](#) - May 2026

(React, TypeScript, ASP.NET Core Web API, Python, PostgreSQL/pgvector, OpenAI API)

- Designed and developed a full-stack AI-powered document intelligence tool enabling users to upload documents and query content through a natural language conversational interface.
- Leveraged generative AI tools (Claude, ChatGPT) during development for prompt design iteration, architecture exploration, and code validation; applied structured review and testing practices to evaluate and verify all AI-assisted output before integration.
- Implemented a RAG (Retrieval-Augmented Generation) pipeline including PDF ingestion, text chunking, vector embedding, and semantic retrieval using pgvector to ground AI responses in source content and reduce hallucination risk.
- Integrated OpenAI API with custom prompt engineering to generate context-aware answers with inline source citations, supporting transparent and verifiable AI output.
- Built a confidence scoring mechanism to flag low-certainty responses for human review, applying responsible AI principles around accuracy, sensitivity, and output validation.
- Implemented streaming response delivery and a user feedback loop to support iterative AI output evaluation and quality monitoring.
- Developed CI/CD pipeline using GitHub Actions with automated unit tests covering AI integration behaviour, retrieval accuracy, and failure handling of AI-driven components.

[Emergency Dispatch Simulation Dashboard](#) - April 2026

(React, .NET Web API, SQL Server, SignalR)

- Designed and developed an enterprise-style operational dispatch management system supporting real-time incident tracking and EMS unit coordination workflows
- Built ASP.NET Core Web API backend services managing incidents, dispatch assignments, audit logging, and unit status workflows following RESTful and object-oriented design principles.
- Implemented SignalR-based real-time communication for live dispatch events, KPI monitoring, and operational dashboard updates.
- Designed normalized SQL Server schemas supporting operational reporting, event tracking, and real-time data integrity.
- Developed React dashboard interfaces supporting incident visibility, operational decision-making, and analytics reporting.
- Integrated lightweight AI-assisted incident analysis logic to recommend priority levels and dispatch unit types based on incident data.

Alberta Road Risk & Infrastructure Intelligence System - May 2026

(React, TypeScript, ASP.NET Core Web API, SQL Server, Python, Leaflet)

- Designed and developed a GIS-enabled operational intelligence platform supporting infrastructure risk analysis and decision-making workflows across Alberta.
- Implemented geospatial mapping and regional risk visualization using Leaflet and GeoJSON, enabling interactive spatial filtering and incident reporting.
- Developed Python ETL workflows for data ingestion, validation, transformation, and normalization supporting analytics processing and risk scoring.
- Designed modular backend services and SQL Server schemas supporting incident management, filtering, and reporting workflows.
- Containerized backend services using Docker to support consistent local development and deployment workflows.

WORK EXPERIENCE

Web Developer

Colombo, Sri Lanka

eBeyonds (Pvt) Ltd | Nov 2020 - Dec 2022, Full-time

- Built and integrated RESTful APIs supporting communication between backend services, frontend applications, and third-party systems, applying object-oriented design principles to improve maintainability and scalability.
- Developed reusable frontend components using JavaScript, HTML, and CSS with responsive design methodologies, accelerating feature delivery across Agile sprint cycles.
- Collaborated with QA teams, business analysts, and stakeholders to translate business requirements into solution design, producing technical specifications and delivering application enhancements through structured sprint cycles.
- Participated in feature estimation, sprint planning, code reviews, and deployment coordination; contributed unit tests and structured test cases to improve release quality and production stability.
- Participated in application troubleshooting, defect resolution, debugging, and performance optimization across multiple environments, producing clear incident reports and status updates for stakeholders.

Trainee Software Engineer

Colombo, Sri Lanka

H One (Pvt) Ltd. | Jan 2020 - Jun 2020, Full-time

- Assisted with backend feature development, debugging, testing, and deployment activities within structured SDLC environments.
- Supported Azure DevOps CI/CD pipeline activities, improving deployment consistency and reducing manual configuration effort.
- Built a PowerApps-based mobile prototype for SriLankan Airlines supporting workflow automation and pilot testing initiatives.
- Collaborated with technical and business stakeholders to translate business requirements into application enhancements and technical solutions, capturing technical and operational requirements.
- Participated in troubleshooting and resolving application issues across development and testing environments, documenting fixes and lessons learned.

Intern – Software Engineer

Colombo, Sri Lanka

Spemai (Pvt) Ltd. | Jun 2019 - Dec 2019, Full-time

- Contributed to backend development for the OnePay digital wallet platform using Python, Django REST APIs, and SQL databases.
- Designed and optimized SQL schemas and queries supporting data integrity and application performance.
- Participated in debugging, issue analysis, API testing, and release support activities
- Collaborated with development teams to support stable software delivery and operational reliability.

File Maintenance Clerk / Operations Support

Edmonton, Alberta

Sobeys | Dec 2022 - Feb 2026, Part-time

- Supported operational pricing, inventory accuracy, and product data maintenance workflows within enterprise retail systems.
- Investigated discrepancies affecting POS pricing, stock accuracy, replenishment workflows, and inventory integrity.
- Worked with systems including SAP, Kronos, ISP Server, POS platforms, and ServiceNow ticketing workflows.
- Assisted with operational troubleshooting and cross-department coordination to support business continuity and issue resolution.

Inventory Associate / Sales Associate (Part-time)

Edmonton, Alberta

Home Depot | Mar 2025 - Present, Part-time

- Supported operational accuracy by identifying inventory discrepancies, contributing to a district-leading shrink rate (~1.47%).
- Performed cycle counts, investigated mismatches, and collaborated with teams to maintain accurate stock data.

EDUCATION

MACEWAN UNIVERSITY (Jan 2023 – Dec 2024)

Post-Graduate Certification in Project Management & Professional Communications | Edmonton, Alberta

Post-Graduate Certification in Full-Stack Web Development & Professional Communications | Edmonton, Alberta

CARDIFF METROPOLITAN UNIVERSITY (Sept 2016 – Nov 2019)

BSc (Hons) in Software Engineering | Wales, United Kingdom *Credential evaluation completed — World Education Services (WES), ECA Report*